

A shopping basket filled with various grocery items like soup, cereal, and snacks, with a person's legs visible in the background.

POTFOLIO PROJECT

CUSTOMERS' PURCHASING PATTERN (CPP) ANALYSIS REPORT

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Introduction

The decision to purchase an item or service by an individual is not often spontaneous but rather due to some form of habit. These behaviours define the purchasing practice of such individual in any sales or commercial transaction with sellers, translating into a model that can be explored and measured as Customers' Purchasing Pattern (CPP). Inclusive of needs, CPP are also influenced by personal, psychological and social traits that are of business interest to enterprise companies who desires information on how customers purchase items, frequency of purchases, customers' product preferences, as well as to plan their marketing strategies and customer engagement processes. In this project, these factors are studied using data analytic tools to generate insights for a business improvement process of an enterprise company.

Goal and Objectives of Study

The overarching goal of the study is to demonstrate the application of data analytic methodology to understanding customers' purchase patterns, and its impact on an improvement process of an enterprise company. The study goal shall be achieved through the following objectives:

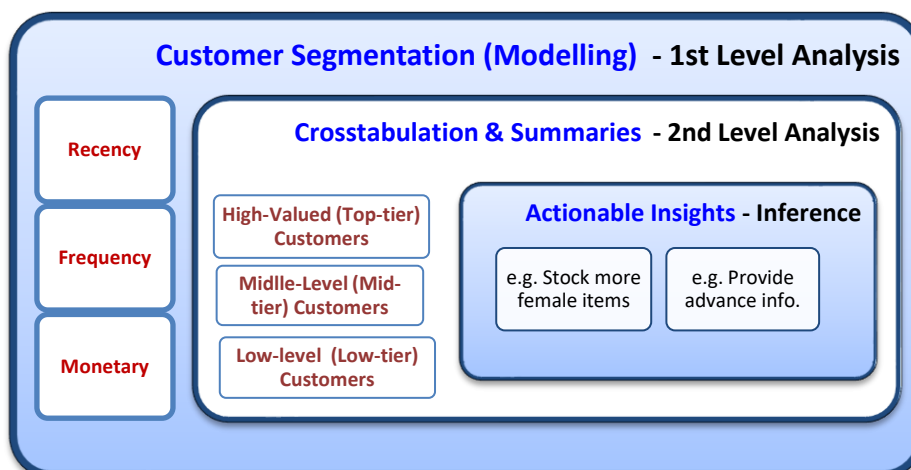
1. Create research questions to guide the study process and to understand the nature of solutions to business issues at hand.
2. Use of a real-life on-line secondary dataset, that would be cleaned and pre-processed for an ideal data analytic task.
3. Write Python codes as a means of implementing technology and algorithmic solutions to solve real-life business issues.
4. Generate insights from the analytical procedure towards understanding the impact of customers' purchase patterns on businesses and based on these findings, make recommendations for consideration by both the C-(management) and operational staff of the enterprise company; and
5. Document the study procedure through the development of a project report targeted at technocrats and PowerPoint-slides for presentation to top management staff of the enterprise company.

Business Questions

1. What are the drivers of item purchases by Customers from the company?
2. Who are the major Customers making purchases from the company?
3. What is the value of item purchased by Customers from the company?
4. How often does an average Customer purchased items from the company?
5. How recent does an average Customers purchased items from the company?

The Analytical Framework

Customers' Purchasing Behaviour cut across several themes including social, economic, technology, demographic, psychology and psychographic factors, as such to perform incisive analyses for actionable insights, the analytical framework must be broad-based and robust. Consequently for the study, an integrated 2-level analytical framework was adopted (see the schema below).



CONTEXTUAL DATA ANALYTICS

Properties of Dataset

The initial dataset structure comprised of 1,966 instances (i.e. a row per customer data items) and 50 features (i.e. columns of data on purchase and other related variables). However, to perform the data-analytic procedures that align effectively with CPP, further feature engineering - specifically feature creation (i.e. creation of new columns of datasets from existing datasets) and feature selection (i.e. selection of column datasets that were best for the CPP analytic process) were done. Consequently 16 more features were created and eventually 28 features were selected to be included in the final CPP analytical procedure using Python in Jupyter – Integrated Development Environment (IDE).

Data Pre-processing

In addition to feature creation and extraction (slicing) specifically for CPP analysis; other data pre-processing tasks were done. These were data cleaning procedures i.e. handling missing values; removing duplicated records and records with blank data-points; detection and treatment of outliers; data, date, and label name formatting; and encoding of categorical variables. Below are some excerpts from the data pre-processing procedures performed.

```
[46]: # Convert all column heading label in a 'snake-structure' format
df.columns=df.columns.str.lower()
df.columns=df.columns.str.replace(' ','_')
#df.columns = df.columns.str.replace(' ','_')
#df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
[47]: df.columns
```

```
[47]: Index(['customer_no', 'most_recent_purchase', 'item_most_recently_purch_1',
        'item_most_recently_purch_2', 'recent_purch_amt_1',
        'recent_purch_amt_2', 'rev_rating_recent_purch',
        'dlr_spent_multi_purch', 'dlr_spent_multi_purch_code', 'order_freq',
        'order_freq_code', 'recency', 'recency_code', 'customer_segment',
        'customer_segment_code', 'yearly_income', 'home_value',
        'subscription_status_1', 'subscription_status_2', 'marital_status',
        'name_prefix', 'age', 'age_category', 'telemarket_ind.',
        'rents_apartment', 'occupied_<1_yn', 'domestic_prod', 'apparel_purch',
        'leisure_prod', 'kitchen_prod', 'luxury_items', 'promo_1-7_mon',
        'promo_8-13_mon', 'county_code', 'total_returns', 'mens_apparel',
        'flatware_purch', 'dishes_purch', 'home_furniture', 'lamps_purch',
        'linens_purch', 'blankets_purch', 'towels_purch', 'outdoor_prod',
        'coats_purch', 'ladies_coats', 'ladies_apparel', 'his_her_apparel',
        'jewelry_purch', 'date_1st_order', 'telemarket_ord', 'account_number',
        'state_code', 'race', 'heating_type', 'number_of_cars',
        'number_of_kids', 'travel_time', 'education_level', 'job_category',
        'dlr_value_per_mailing', 'total_dining_kitch_dish_flat', 'sex',
        'sex_dummy_variable', 'men_-_purchase', 'women_purchase'],
        dtype='object')
```

```
[51]: # To create a new column 'Age-group' in the dataframe
```

```
labels = ['Adolescent','Youth','Young Adult','Adult']
df['age_group']=pd.qcut(df['age'],q=4,labels=labels)
#print(df[['age_category','age_group']])
# OR
df1=df[['age_category','age_group','sex']]
df1.head()
```

```
[51]:
```

	age_category	age_group	sex
0	Youth	Young Adult	female
1	Youth	Adolescent	female
2	Youth	Adult	male
3	Adult	Adult	female
4	Youth	Adolescent	female

```
[56]: # Extracting 28 features required for the CPP Analytic Procedure
dfcpp=df.iloc[:,[0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,17,18,19,21,22,53,58,59,60,61,62,63,64,65,66]]
dfcpp.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1966 entries, 0 to 1965
Data columns (total 31 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   customer_no                          1966 non-null   int64
1   most_recent_purchase                 1966 non-null   int64
2   item_most_recently_purch_1          1966 non-null   object
3   item_most_recently_purch_2          1966 non-null   object
4   recent_purch_amt_1                   1966 non-null   int64
5   recent_purch_amt_2                   1966 non-null   int64
6   rev_rating_recent_purch              1966 non-null   float64
7   dlr_spent_multi_purch                 1966 non-null   int64
8   dlr_spent_multi_purch_code           1966 non-null   int64
9   order_freq                           1966 non-null   float64
10  order_freq_code                       1966 non-null   int64
11  recency                              1966 non-null   int64
12  recency_code                          1966 non-null   int64
```

```
13  customer_segment                     1966 non-null   object
14  customer_segment_code                 1966 non-null   int64
15  yearly_income                         1966 non-null   int64
16  subscription_status_1                 1966 non-null   int64
17  subscription_status_2                 1966 non-null   object
18  marital_status                       1966 non-null   int64
19  age                                   1966 non-null   int64
20  age_category                          1966 non-null   object
21  race                                  1966 non-null   int64
22  education_level                       1966 non-null   int64
23  job_category                          1966 non-null   int64
24  dlr_value_per_mailing                  1966 non-null   float64
25  total_dining_kitch_dish_flat          1966 non-null   int64
26  sex                                    1966 non-null   object
27  sex_dummy_variable                    1966 non-null   int64
28  men_purchase                          1966 non-null   int64
29  women_purchase                        1966 non-null   int64
30  age_group                             1966 non-null   category
```

```
[63]: # Extracting (slicing) the MRF features from the main dataset
```

```
#dfcpp_mrf=df.iloc[:,[7,9,11]]
#dfcpp_mrf.columns, dfcpp_mrf.shape

dfcpp_mrf_anal=df.iloc[:,[0,5,7,9,11,14,21,62,66]]
dfcpp_mrf_anal.columns, dfcpp_mrf_anal.shape
```

```
[63]: (Index(['customer_no', 'recent_purch_amt_2', 'dlr_spent_multi_purch',
            'order_freq', 'recency', 'customer_segment_code', 'age', 'sex',
            'age_group'],
           dtype='object'),
      (1966, 9))
```

```
[71]: # Extracting the 'Premium Customer' dataframe only from both 'MRF & Customer_segment_code'
# and the main dataframes (i.e df)
```

```
prem_cust=dfcpp_mrf_anal.loc[dfcpp_mrf_anal['customer_segment_code']==3]
prem_cust.columns, prem_cust.shape, prem_cust.head(200)
```

```
[71]: (Index(['customer_no', 'recent_purch_amt_2', 'dlr_spent_multi_purch',
            'order_freq', 'recency', 'customer_segment_code', 'age', 'sex',
            'age_group'],
           dtype='object'),
      (109, 9),
      customer_no      recent_purch_amt_2  dlr_spent_multi_purch  order_freq \
37          38              0              3553          9.43
243         244             100             1293          16.21
282         283              94             1158          12.96
297         298              26             6367           6.37
308         309              29             2149          10.26
...         ...              ...              ...          ...
1948        1949              37              342          10.00
1950        1951              78              618          16.32
1952        1953              82              839          16.32
1954        1955              77              775          16.32
1955        1956              0              531          10.20

      recency  customer_segment_code  age  sex  age_group
37          7                    3   34  female  Young Adult
243         0                    3   61   male    Adult
282        17                    3   30  female    Youth
297         1                    3   33  female  Young Adult
308        45                    3   42   male    Adult
...         ...              ...   ...   ...   ...
1948        15                    3   32   male  Young Adult
1950        15                    3   29  female    Youth
1952        16                    3   29  female    Youth
1954        51                    3   44  female    Adult
1955         8                    3   36  female    Adult

[109 rows x 9 columns])
```

```
[72]: # Extracting the 'Opportunity Customer' dataframe only from both 'MRF & Customer_segment_code'
# and the main dataframes (i.e df)
```

```
opp_cust=dfcpp_mrf_anal.loc[dfcpp_mrf_anal['customer_segment_code']==2]
opp_cust.columns, opp_cust.shape, opp_cust.head(1000)
```

```
[72]: (Index(['customer_no', 'recent_purch_amt_2', 'dlr_spent_multi_purch',
'order_freq', 'recency', 'customer_segment_code', 'age', 'sex',
'age_group'],
dtype='object'),
(933, 9),
customer_no recent_purch_amt_2 dlr_spent_multi_purch order_freq \
0 1 53 740 1.23
1 2 0 333 3.62
2 3 0 208 1.32
3 4 90 1840 7.28
4 5 0 325 0.26
...
1958 1959 67 220 4.08
1959 1960 0 533 8.16
1961 1962 0 419 6.12
1963 1964 30 291 4.08
1964 1965 51 575 6.12

recency customer_segment_code age sex age_group
0 0 2 31 female Young Adult
1 4 2 25 female Adolescent
2 2 2 36 male Adult
3 0 2 41 female Adult
4 71 2 26 female Adolescent
...
1958 33 2 25 female Adolescent
1959 7 2 29 female Youth
1961 36 2 36 male Adult
1963 33 2 33 female Young Adult
1964 49 2 21 male Adolescent

[933 rows x 9 columns])
```

```
[73]: # Extracting the 'Least Customer' dataframe only from both 'MRF & Customer_segment_code'
# and the main dataframes (i.e df)
```

```
lst_buy_cust=dfcpp_mrf_anal.loc[dfcpp_mrf_anal['customer_segment_code']==1]
lst_buy_cust.columns, lst_buy_cust.shape, lst_buy_cust.head(1000)
```

```
[73]: (Index(['customer_no', 'recent_purch_amt_2', 'dlr_spent_multi_purch',
'order_freq', 'recency', 'customer_segment_code', 'age', 'sex',
'age_group'],
dtype='object'),
(924, 9),
customer_no recent_purch_amt_2 dlr_spent_multi_purch order_freq \
5 6 20 0 5.70
6 7 85 44 1.33
10 11 34 0 1.08
11 12 0 0 0.79
12 13 0 0 1.34
...
1944 1945 45 44 2.00
1951 1952 0 152 4.08
1960 1961 72 98 4.08
1962 1963 45 79 4.08
1965 1966 90 124 4.08

recency customer_segment_code age sex age_group
5 559 1 25 female Adolescent
6 80 1 26 female Adolescent
10 615 1 41 female Adult
11 364 1 30 female Youth
12 563 1 32 male Young Adult
...
1944 183 1 26 female Adolescent
1951 96 1 44 female Adult
1960 37 1 31 female Young Adult
1962 91 1 38 female Adult
1965 33 1 42 female Adult

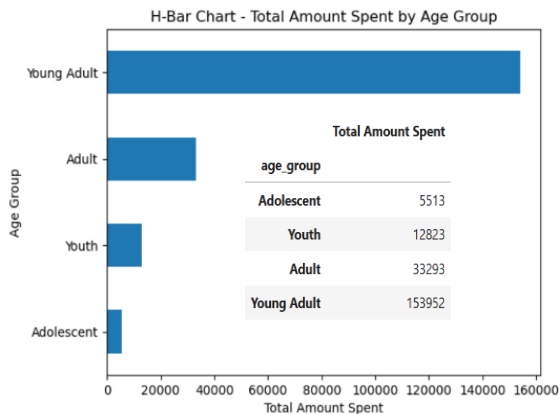
[924 rows x 9 columns])
```

Exploratory Data Analysis (EDA) of the MFR Dataset

Before performing the basic data analyses, EDA procedures were performed, to examining the structure within the MFR dataset and to summarize the relationships in this dataset. The procedure analysed the identified three (3) main types of 'Customer Segments' (Customer segmentation – Top-tier [Premium], Mid-tier [Opportunity], & Low-tier [Low-buying] Customers) in the MRF dataset. The characteristics of each customer types were further explored by age, age-group and sex of customers. The results were presented in tables, and vertical-bar, horizontal-bar & pie charts. Below are some excerpts from the EDA procedures by customer type.

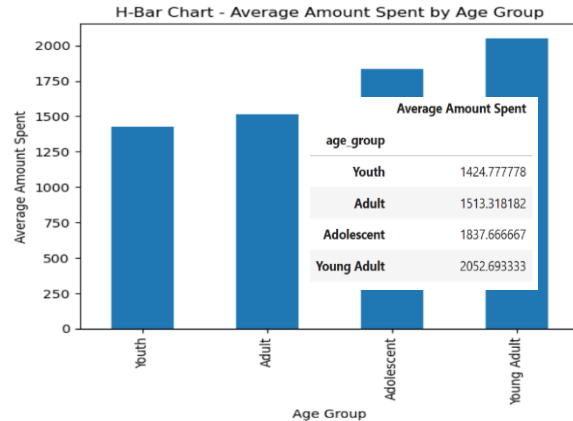
TOP-TIER CUSTOMERS

'Premium Customers Previous Purchases'



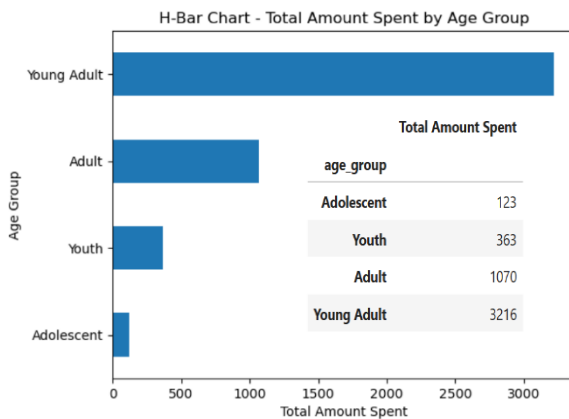
Finding: By Age-group and total amount spent overtime, in the "Top-tier Customers" category, Young Adults (25-35) spent significantly most (more than others combined), followed by Adults (36-65), while Adolescents (10-15) spent the least. **Insight:** Reward and customer loyalty processes should be implemented, high-valued and age-range items should always be stocked, to target and further encourage Young Adult customers to maintain their spending patterns, while Adults, Youths (16-24) and Adolescents should be targeted with sales promotion strategies to improve their spending pattern on the Company's items.

'Premium Customers Previous Purchases'



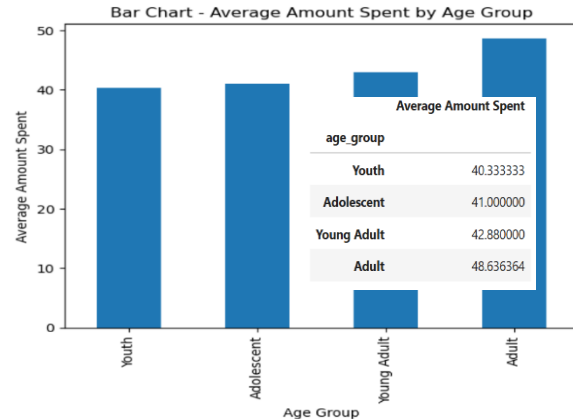
Finding: By Age-group and on average amount spent to previously purchase items, in the "Top-tier Customers" category, Young Adults and Adolescents spent more with an almost equal average amount spent in both age categories. Adults and Youths spent less with also almost equal average amount spent in both age categories. **Insight:** All customers (especially Adults and Youths) must be encouraged with top-notch customer relation and sale improvement strategies, to maintain continuously and even increase their spending levels on the Company's items

'Premium Customers Current Purchase'



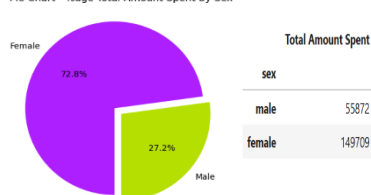
Finding: By Age-group and total amount spent on currently purchased items, in the "Top-tier Customers" category, Young Adults spent significantly more than all other customer age-group categories combined, followed by Adults, while Adolescents spent the least. **Insight:** Reward and customer loyalty processes should be implemented, high-valued and age-relevant items should always be in stock, to target and further encourage both Young Adult customers to maintain their spending pattern, while Adults, Youths and Adolescents should be targeted with sales promotion strategies to improve their spending pattern on the Company's items.

'Premium Customers Current Purchase'



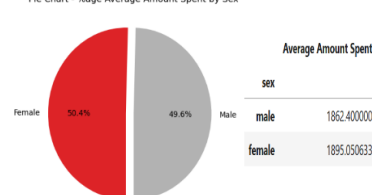
Finding: By Age-group and average amount spent to currently purchase the Company's items, in the "Top-tier Customers" category, all customers in the different age-groups spent almost equally, with Adults spending marginally more than others. **Insight:** All customers must be encouraged with top-notch customer relation and sale improvement strategies to maintain continuously and even increase their spending levels on the Company's items.

'Premium Customers Previous Purchases':
Pie Chart - %Age Total Amount Spent by Sex



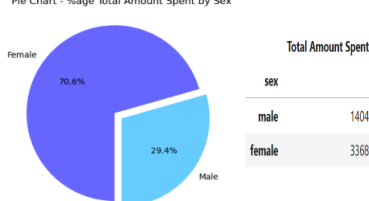
Finding: Female customers dominates both in size and total amount spent on previously purchased items by "Top-tier Customers" of the Company. **Insight:** Reward and customer loyalty processes should be implemented to further encourage female "Top-tier Customers", to continue patronizing the Company, while sales campaigns to target increasing the number of male customers specifically should be implemented.

'Premium Customers Previous Purchases':
Pie Chart - %Age Average Amount Spent by Sex



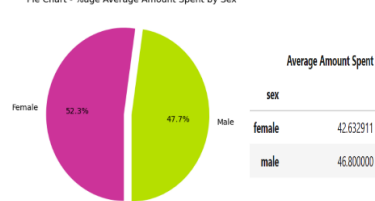
Finding: Both female and male customers almost spent equally on the average on previously purchased items by "Top-tier Customers" of the Company, with female customers average spending marginally higher than male customers. **Insight:** All customers (both male and female) should be encouraged with customer relation and sale strategies to maintain or even increase their spending levels on the Company's items.

'Premium Customers Current Purchase':
Pie Chart - %Age Total Amount Spent by Sex



Finding: Female customers dominates both in size and total amount spent on currently purchased items by "Top-tier Customers" of the Company. **Insight:** Reward and customer loyalty processes should be implemented to further encourage female "Top-tier Customers", to continue patronizing the Company, while sales campaigns targeting increasing the number of male customers specifically should be implemented.

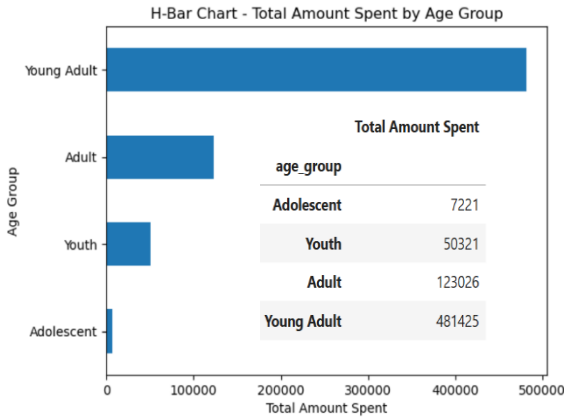
'Premium Customers Current Purchase':
Pie Chart - %Age Average Amount Spent by Sex



Finding: Both female and male customers almost spent equally on the average on currently purchased items by "Top-tier Customers" of the Company, with female customers average spending marginally higher than that of male customers. **Insight:** All customers (both male and female) should be encouraged with customer relation and sale strategies to maintain or even increase their spending levels on the Company's items

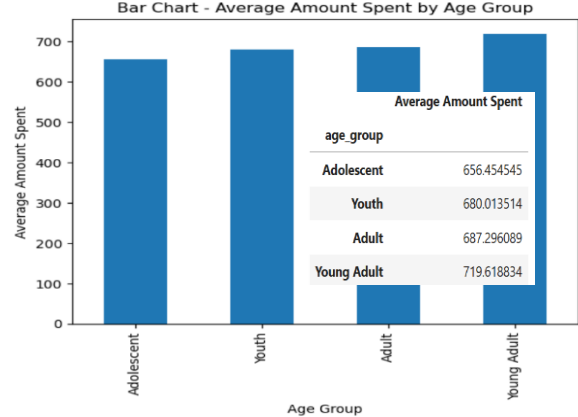
MID-TIER CUSTOMERS

'Opportunity Customers Previous Purchases'



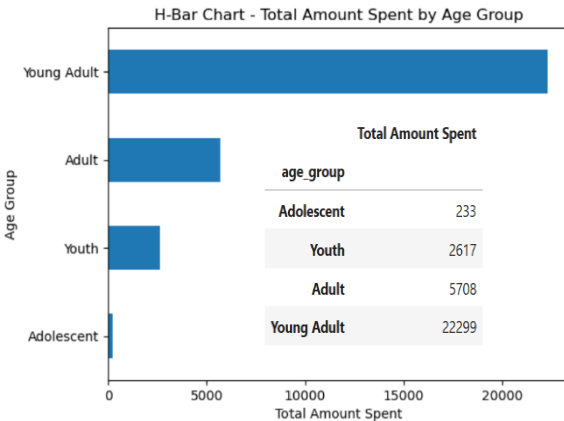
Finding: By Age-group and total amount spent overtime, in the "Mid-tier Customers" category, Young Adults spent most (significantly more than all other customer's age-groups combined), followed by Adults, while Youths spent the least. **Insight:** Reward and customer loyalty processes should be implemented, high-valued and age-range items should always be stocked, to target and further encourage Young Adult customers to maintain their spending patterns, while Adults, Youths and Adolescents should be targeted with sales promotion strategies to improve their spending pattern on the Company's items.

'Opportunity Customers Previous Purchases'



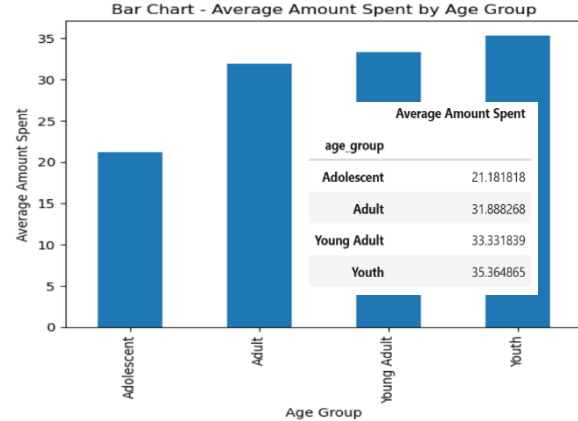
Finding: By Age-group and average amount spent on previously purchased items, in the "Mid-tier Customers" category, all customers in the different age-groups spent almost equally, with Young Adults marginally spending more than others. **Insight:** All customers must be encouraged with top-notch customer relation and sale improvement strategies to maintain continuously and even increase their spending levels on the Company's items.

'Opportunity Customers Current Purchase'



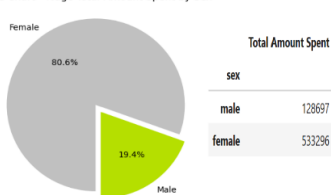
Finding: By Age-group and total amount spent on currently purchased items, in the "Mid-tier Customers" category, Young Adults spent most (significantly more than all other customer's age-groups combined), followed by Adults, while Adolescents spent the least. **Insight:** Reward and customer loyalty processes should be implemented, high-valued and age-relevant items should always be in stock, to target and further encourage both Young Adult customers to maintain their spending patterns, while Adults, Youths and Adolescents should be targeted with sales promotion strategies to improve their spending pattern on the Company's items.

'Opportunity Customers Current Purchase'



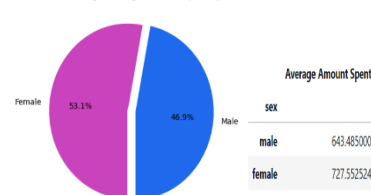
Finding: By Age-group and on average amount spent to currently purchase items, in the "Mid-tier Customers" category, Youths spent most, followed by Young Adults and Adults that both spent almost equally, while Adolescents spent least. **Insight:** All customers (especially Young Adults and Adults) must be encouraged with top-notch customer relation and sale improvement strategies to maintain continuously and even increase their spending levels on the Company's items, while Adolescent customers should be targeted with more age-range items and sale strategies that will improve on their average spending on these items.

'Opportunity Customers Previous Purchases':
Pie Chart - %age Total Amount Spent by Sex



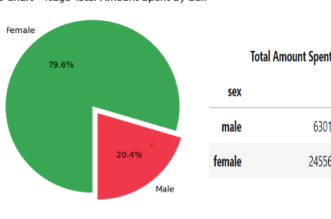
Finding: Female customers dominates both in size and total amount spent on previously purchased items by "Mid-tier Customers" of the Company. **Insight:** Reward, customer loyalty, and sale improvement processes should be implemented to further encourage female "Mid-tier Customers", to continue patronizing the Company, while sales campaigns to increase the number of male customers specifically should be implemented.

'Opportunity Customers Previous Purchases':
Pie Chart - %age Average Amount Spent by Sex



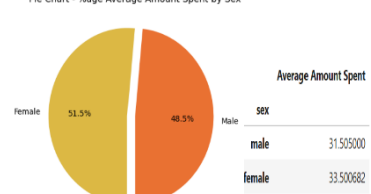
Finding: Both female and male customers almost spent equally on the average on previously purchased items by "Mid-tier Customers" of the Company, with female customers average spending marginally higher than male customers. **Insight:** All customers (both male and female) should be encouraged with customer relation and sale strategies to increase their spending levels on the Company's items and possibly step up to become "Top-tier Customers"

'Opportunity Customers Current Purchase':
Pie Chart - %age Total Amount Spent by Sex



Finding: Female customers dominates both in size and total amount spent on currently purchased items by "Mid-tier Customers" of the Company. **Insight:** Reward, customer loyalty, and sale improvement processes should be implemented to further encourage female "Mid-tier Customers", to continue patronizing the Company, while sales campaigns to increase the number of male customers specifically should be implemented.

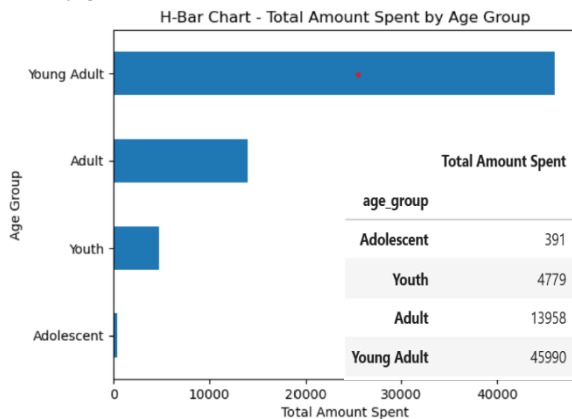
'Opportunity Customers Current Purchase':
Pie Chart - %age Average Amount Spent by Sex



Finding: Both female and male customers almost spent equally on the average on currently purchased items by "Mid-tier Customers" of the Company, with female customers average spending marginally higher than male customers. **Insight:** All customers (both male and female) should be encouraged with customer relation and sale strategies to increase their spending levels on the Company's items and possibly step up to become "Top-tier Customers"

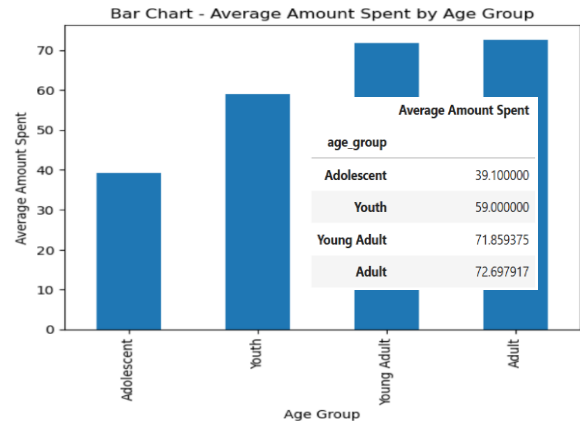
LOW-TIER CUSTOMERS

'Least Buying Customers Previous Purchases'



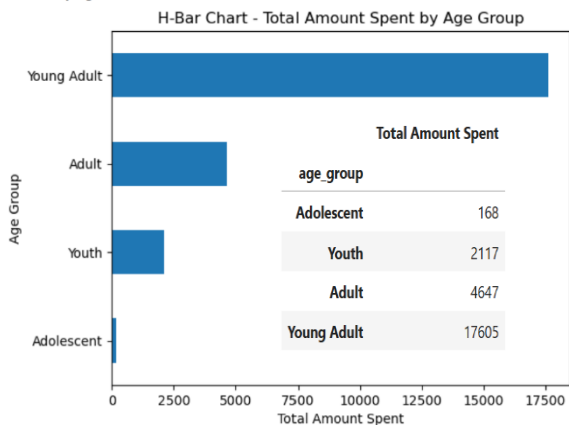
Finding: By Age-group and total amount spent overtime, in the "Low-tier Customers" category, Young Adults spent most (significantly more than all other customer's age-groups combined), followed by Adults, while Adolescents spent the least. **Insight:** Reward and customer loyalty processes should be implemented, high-valued and age-relevant items should always be stocked, to target and further encourage Young Adult customers to maintain their spending pattern, while Adults, Youths and Adolescents should be targeted with sales promotion strategies to improve their spending pattern on the Company's items.

'Least Buying Customers Previous Purchases'



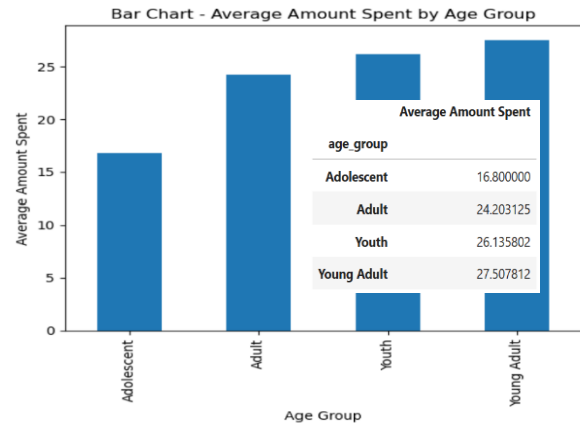
Finding: By Age-group and average amount spent on previously purchased items, in the "Low-tier Customers" category, Adult and Young Adult customers spent most (almost equally), followed by Youths, while Adolescents spent least. **Insight:** All customers (especially Adolescents) must be encouraged with top-notch customer relation and sale improvement strategies to continuously maintain and even increase their spending levels on the Company's items.

'Least Buying Customers Current Purchase'



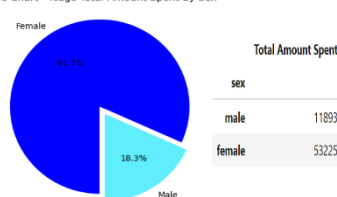
Finding: By Age-group and total amount spent to currently purchase items, in the "Low-tier Customers" category, Young Adults spent most (significantly more than all other customer's age-groups combined), followed by Adults, while Adolescents spent the least. **Insight:** Reward and customer loyalty processes should be implemented, high-valued and age-relevant items should always be stocked, to target and further encourage Young Adult customers to maintain their spending pattern, while Adults, Youths and Adolescents should be targeted with sales promotion strategies to improve their spending pattern on the Company's items.

'Least Buying Customers Current Purchase'



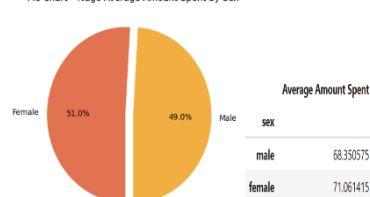
Finding: By Age-group and average amount spent on currently purchased items, in the "Low-tier Customers" category, Young Adult and Youth customers spent most (almost equally), followed by Adults, while Adolescents spent least. **Insight:** All customers (especially Adolescents) must be encouraged with top-notch customer relation and sale improvement strategies to continuously maintain and even increase their spending levels on the Company's items.

'Least Buying Customers Previous Purchases:'
Pie Chart - %Age Total Amount Spent by Sex



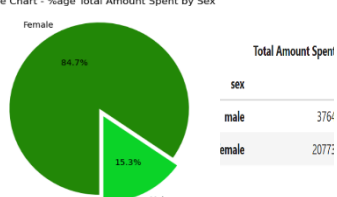
Finding: Female customers dominates both in size and total amount spent on previously purchased items by "Low-tier Customers" of the Company. **Insight:** Reward, customer loyalty, and sale improvement processes should be implemented to further encourage female "Low-tier Customers", to patronize the Company more, while sales campaigns to increase the number of male customers specifically should be implemented.

'Least Buying Customers Previous Purchases:'
Pie Chart - %Age Average Amount Spent by Sex



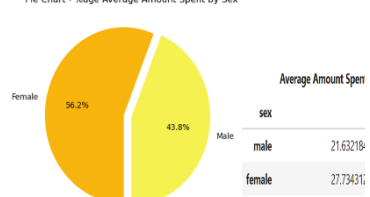
Finding: Both female and male customers almost spent equally on the average on previously purchased items by "Low-tier Customers" of the Company, with female customers average spending marginally higher than male customers. **Insight:** All customers (both male and female) should be encouraged with customer relation and sale strategies to improve their spending levels on the Company's items and possibly step up to become "Mid-tier Customers" or even "Top-tier Customers".

'Least Buying Customers Current Purchase:'
Pie Chart - %Age Total Amount Spent by Sex



Finding: Female customers dominates both in size and total amount spent on currently purchased items by "Low-tier Customers" of the Company. **Insight:** Reward, customer loyalty, and sale improvement processes should be implemented to further encourage female "Low-tier Customers", to patronize the Company more, while sales campaigns to increase the number of male customers specifically should be implemented.

'Least Buying Customers Current Purchase:'
Pie Chart - %Age Average Amount Spent by Sex

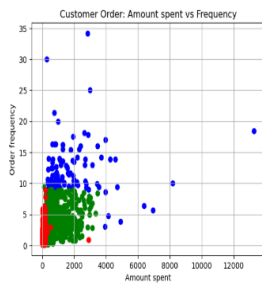


Finding: Both female and male customers almost spent equally on the average on currently purchased items by "Low-tier Customers" of the Company, with female customers average spending marginally higher than male customers. **Insight:** All customers (both male and female) should be encouraged with customer relation and sale strategies to improve their spending levels on the Company's items and possibly step up to become "Mid-tier Customers" or even "Top-tier Customers".

The EDA process also explore relationships between the MFR dataset i.e. the analyses of bivariate cross-plots of Monetary, Frequency and Recency data points across all the identified Customer segments. The findings and insights drawn are presented below.

MFR Dataset Interaction (Scatterplot)

Monetary vs Frequency



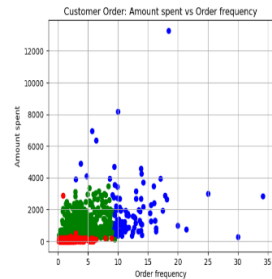
Finding: "Top-tier Customers" spent the most money on purchases and also purchased items most frequently from the Company. **Insight:** Items most purchases and/or frequently ordered by "Top-tier Customers", should be identified, always stocked and made available to this category of customers, as well as conduct of targeted promotion of other less purchased items to encourage them to buy more.

Finding: "Mid-tier Customers" spent less on purchases and also ordered less frequently than "Top-tier Customers" but spent more and ordered more frequently than "Low-tier Customers" from the Company. **Insight:**

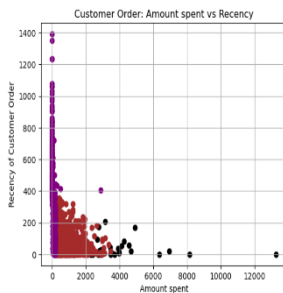
"Mid-tier Customers" should be targeted with sale strategies

and promotions, on all items sold by the Company to encourage them to buy more order more frequently, and if possible, transit to become "Top-tier Customers".

Finding: "Low-tier Customers" spent the least on purchases and also purchased items least frequently from the Company. **Insight:** A Customer satisfaction opinion poll should be conducted among "Low-tier Customers" to understand the rationale for their low spending power and patronage. Identified issues should be addressed with appropriate solutions including targeted interventions that can increase their spending strategies and frequency of buying the Company's items.



Monetary vs Recency



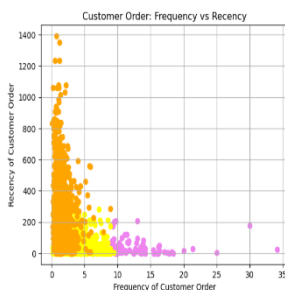
Finding: "Top-tier Customers" spent the most money on purchases and also most recently purchase items from the Company. **Insight:** The company should identify and sustain all their sale strategies that satisfied "Top-tier Customers" to further encourage them to spend more and continue to order items within the shortest time interval. Also in a Customer Satisfaction study, "Top-tier Customers" should be asked to suggest other ways through which the Company can serve them better now and in future to further boost their spend and shorten the time interval of their orders.

Finding: "Mid-tier Customers" spent less on purchases and also order less recently than "Top-tier Customers". but spent more and order more recently than "Low-tier Customers" from the Company. **Insight:** "Mid-tier Customers" should be targeted with sale strategies and

promotions, on all items sold by the Company to encourage them to spend more, further shorten their item order intervals.

Finding: "Low-tier Customers" spent the least on purchases and also least recently (longest time intervals between) item purchases from the Company. **Insight:** A Customer satisfaction opinion poll should be conducted among "Low-tier Customers" to understand the rationale for their low spending power and patronage. Identified issues should be addressed with appropriate solutions including targeted interventions, that can boost their spending power and shorten the time intervals of ordering items from the Company.

Frequency vs Recency



Finding: "Top-tier Customers" bought items most frequently and most recently ordered items from the Company.

Insight: Reward and customer loyalty processes should be implemented to further encourage "Top-tier Customers", to continue to frequently order items and sustain the short time interval they order items from the Company.

Finding: "Mid-tier Customers" less frequently purchased items and also less recently ordered items than "Top-tier Customers" from the Company, but more frequently bought and ordered items more recently than "Low-tier Customers" from the Company. **Insight:** "Mid-tier Customers" should be targeted with sale strategies and promotions, on all items sold by the Company to encourage them to buy more frequently and shorten the time

interval of ordering items from the Company.

Finding: "Low-tier Customers" least frequently purchase items and also least recently (took longer time intervals to) order items from the Company. **Insight:** A Customer satisfaction opinion poll should be conducted among "Low-tier Customers" to understand rationale for their less patronage of the Company's items and why they long time intervals between purchases. Identified issues should be addressed with appropriate solutions including targeted interventions to increase there patronage of the Company's items and reduce significantly the time intervals of their purchases.

CPP ANALYSIS – BUSINESS INTELLIGENCE INSIGHTS FROM SQL DATABASE PROCEDURE

The following business data supported inquiries were evaluated to provide insights for the Company's management, towards further understanding of the purchasing pattern of customers, and thus plan sales strategies for the different customer - segments, sex, and age-group.

1. Revenue earned by the Company from previous sales to Customers, by sex category:

	Sex	Total_Revenue
▶	female	736354
	male	196462

2. Number of Customers earning above \$100,000 yearly, and whose total previous spending to purchase items from the Company was above the Company's all-Customer's average spending on previously purchase items.

No of Customer with Annual Income > \$100,000 & Total previous spending > All-Customer's average previous spending on Company's items
576

3. Number of Customers with Oder frequency more than four (4), and whose total previous spending to purchase items from the Company was above the Company's all-Customer's average spending on previously purchase items.

No of Customer with Order frequency >= 5 & Total previous spending > All-Customer's average previous spending on Company's items
249

4. No of Customer whose recent item purchase was two (2) years or less and total previous spending to purchase items from the Company was above the Company's all-Customer's average spending on previously purchase items.

No of Customer with whose recent item purchase was <= 2 years & Total previous spending > All-Customer's average previous spending on Company's items
575

5. List of ten (10) items recently bought from the Company - by item name, total amount earned from purchased items, and the average rating of each item presented in descending order.

	Recently Purchased Item	Total Amount of Recently Purchased Item	Average Product Rating
▶	Dress	1872	3.98
	Sneakers	2409	3.9
	Coat	3166	3.89
	Sweater	2528	3.87
	Belt	2357	3.84
	Shorts	2853	3.83
	Skirt	2322	3.83
	Blouse	2201	3.78
	Pants	2957	3.78
	Socks	1945	3.77

6. Classifying Customers by their age-group, total number of Customers in each age group in descending order and the average amount spent to previously purchase items from the Company by Customers in each age-group (or category).

Age Group	No of Customers by Age Category	Average Amount Spent
▶ Young Adult	1384	492.32
Adult	394	432.49
Youth	164	414.16
Adolescent	24	546.88

7. Classifying Customers by their subscription status, total number of Customers in each subscription status, the average amount and the total amount (in descending order) spent on previously purchased items from the Company by Customers in each subscription status category.

Customer subscription status	No of Customers	Average amount spent on previous item purchases	Total amount spent on previous item purchases
▶ No	993	458.23	455022
Yes	973	491.05	477794

8. Classifying Customers by their customer segment, and the total number of Customers in each segment.

Customer Segment	Total Customer in each Segment
▶ Top-tier	109
Low-tier	924
Mid-tier	933

9. Within each customer segment, ranking in descending order, the most three (3) items currently purchased by Customers from the Company, in terms of total amount in sales.

Item_rank	Customer_segment	Three_most_recently_purchased_items	Total_amount_of_recently_purchased_items
▶ 1	Low-tier	Coat	1872
2	Low-tier	Boots	1396
3	Low-tier	Jacket	1280
1	Mid-tier	Hoodie	1631
2	Mid-tier	Pants	1621
3	Mid-tier	Boots	1617
1	Top-tier	Shorts	351
2	Top-tier	Sweater	347
3	Top-tier	Shoes	335

10. Classifying 'most frequent buying' Customers of the Company, according to their subscription status.

Subscription status	Most frequent buyers
▶ Yes	47
No	53

10. Classifying Customers by their age-group and the total amount spent in descending order, to previously purchase items from the Company.

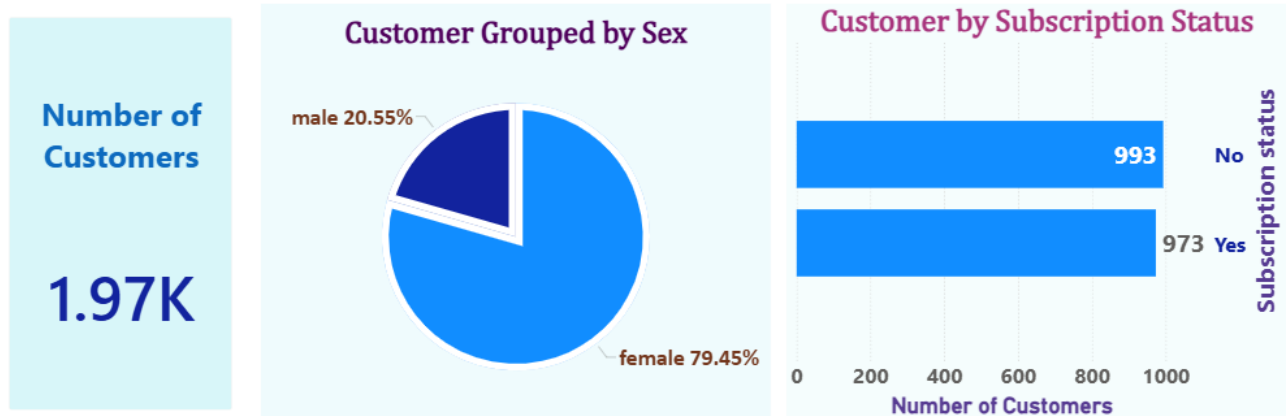
	Customer age group	Total amount from previously sold items
▶	Young Adult	681367
	Adult	170401
	Youth	67923
	Adolescent	13125

11. Classify Customers by their age-group and the total amount spent in descending order, on currently purchased items from the Company.

	Customer age group	Total amount from recently purchased items
▶	Adolescent	524
	Youth	5097
	Adult	11515
	Young Adult	43120

CPP ANALYSIS – BUSINESS INTELLIGENCE DASHBOARD

METRICS & CHARTS - CUSTOMER CATETORIES



Currently Purchased Items' Metrics

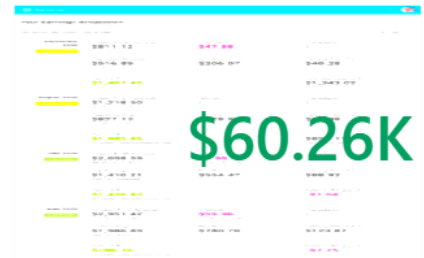
No of Customers that Currently Bought Items



Average Amount Spent



Total Amount Spent



TABLES & CHARTS - CURRENTLY PURCHASED ITEMS

Chart: Top three (3) recently purchased items by Total amount of recently purchased items

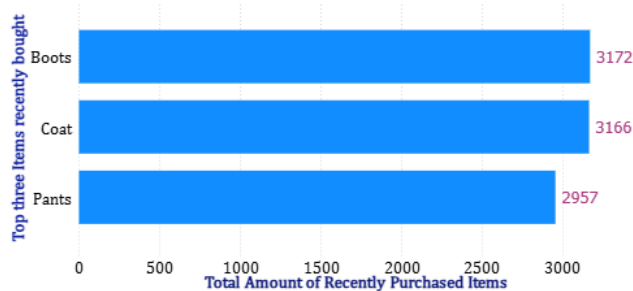


Table: Top three (3) items recently bought

item_most_recently_purch_2	Total Amount of Recently Purchased Items	Item Rank by Amount
Boots	3172	1
Coat	3166	2
Pants	2957	3

Chart: Total amount of recently purchased items by Least three (3) recently purchased items

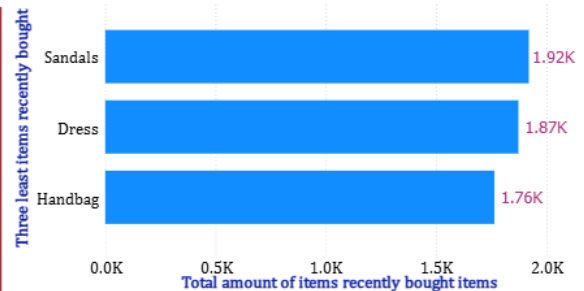


Table: Least three (3) items recently bought

item_most_recently_purch_2	Total Amount of Recently Purchased Items	Item Rank by Amount
Sandals	1919	23
Dress	1872	24
Handbag	1763	25

METRICS & CHARTS - PREVIOUS - ALL TIME ITEM PURCHASES

Amount Spent on Items Previously Purchased

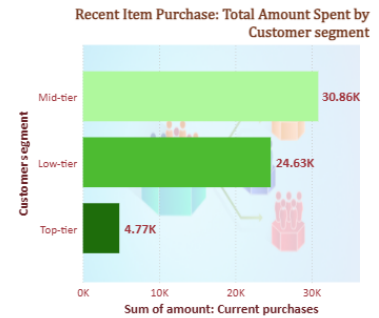
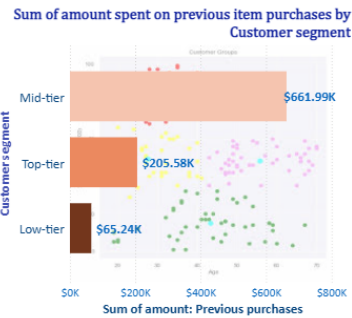
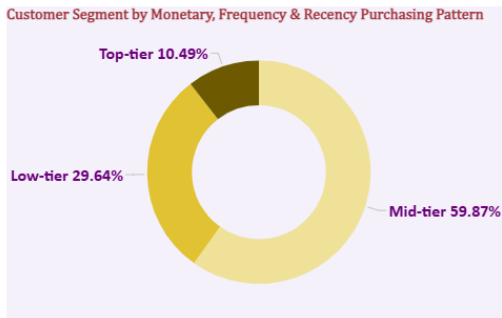
Average Amount Spent

\$474.47

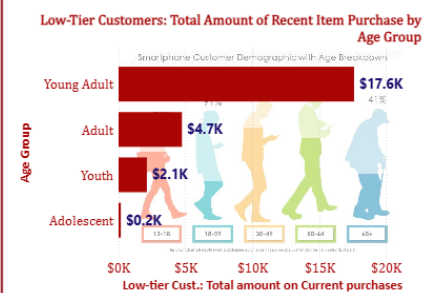
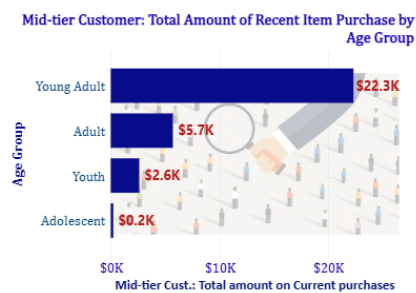
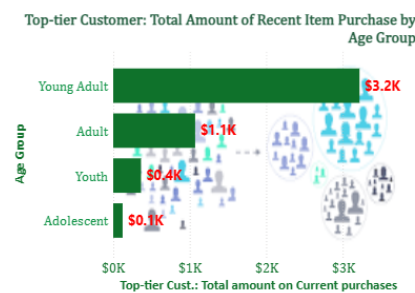
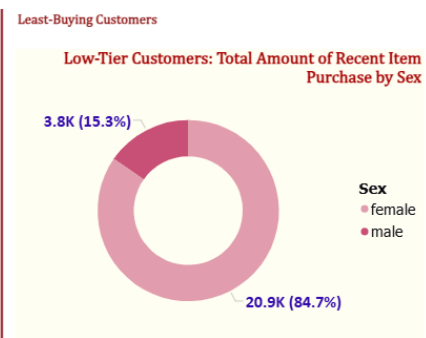
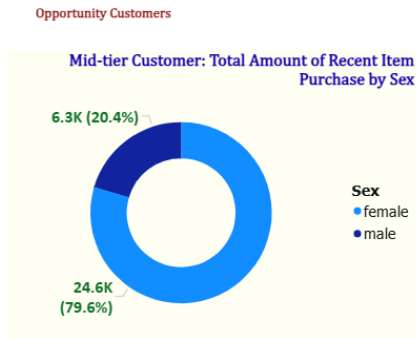
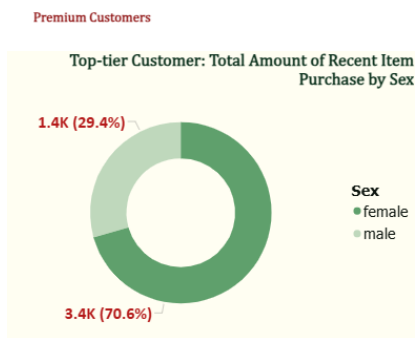
Total Amount Spent

\$932.82K

CHARTS - CUSTOMER SEGMENT ANALYSES



CHARTS & TABLES - CUSTOMER SEGMENT CATEGORY ANALYSES



Top-Tier Customers: Top Three (3) Items Currently Purchased

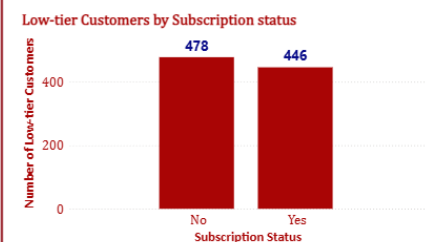
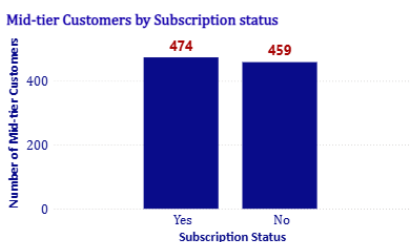
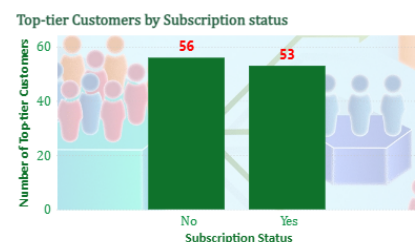
item_most_recently_purch_2	Top-tier Customer: Total Amount of Recent Item Purchase	Top-tier Customer: Item Rank by Amount (Within Cust Segment)
Shorts	351	1
Sweater	347	2
Shoes	335	3

Mid-Tier Customers: Top Three (3) Items Currently Purchased

item_most_recently_purch_2	Mid-tier Customer: Total Amount of Recent Item Purchase	Mid-tier Customer: Item Rank by Amount (Within Cust Segment)
Hoodie	1631	1
Pants	1621	2
Boots	1617	3

Low-Tier Customers: Top Three (3) Items Currently Purchased

item_most_recently_purch_2	Low-Tier Customers: Total Amount of Recent Item Purchase	Low-Tier Customers: Item Rank by Amount (Within Cust Segment)
Coat	1872	1
Boots	1396	2
Jacket	1280	3



Conclusion

CPP was defined as the purchasing habit of buyers, which is greatly influenced by personal trait, demographic, and psychographic factors among others. The MRF analytic framework adopted to explore purchasing behaviour did well to provide basis for three identified categories of customers. Within each customer segment different purchasing behaviours were evident, and insights generated accordingly. Basically the study explored CPP from the perspective of businesses, understanding their customers better and using this relationship to promote sales improvement processes and other business management strategies. Looking forward, customers will continue to drive sale successes and businesses that understand this, would do well to invest in empowering their customers with top-notch customer services, MFR knowledge and feedback insights, and most essentially best resource longer-mile benefits that would align their purchasing patterns with the strategic thinking of the company.

APPENDIX A

DATA SOURCE: <https://www.kaggle.com/brijbhushannanda1979/bigmart-sales-data>

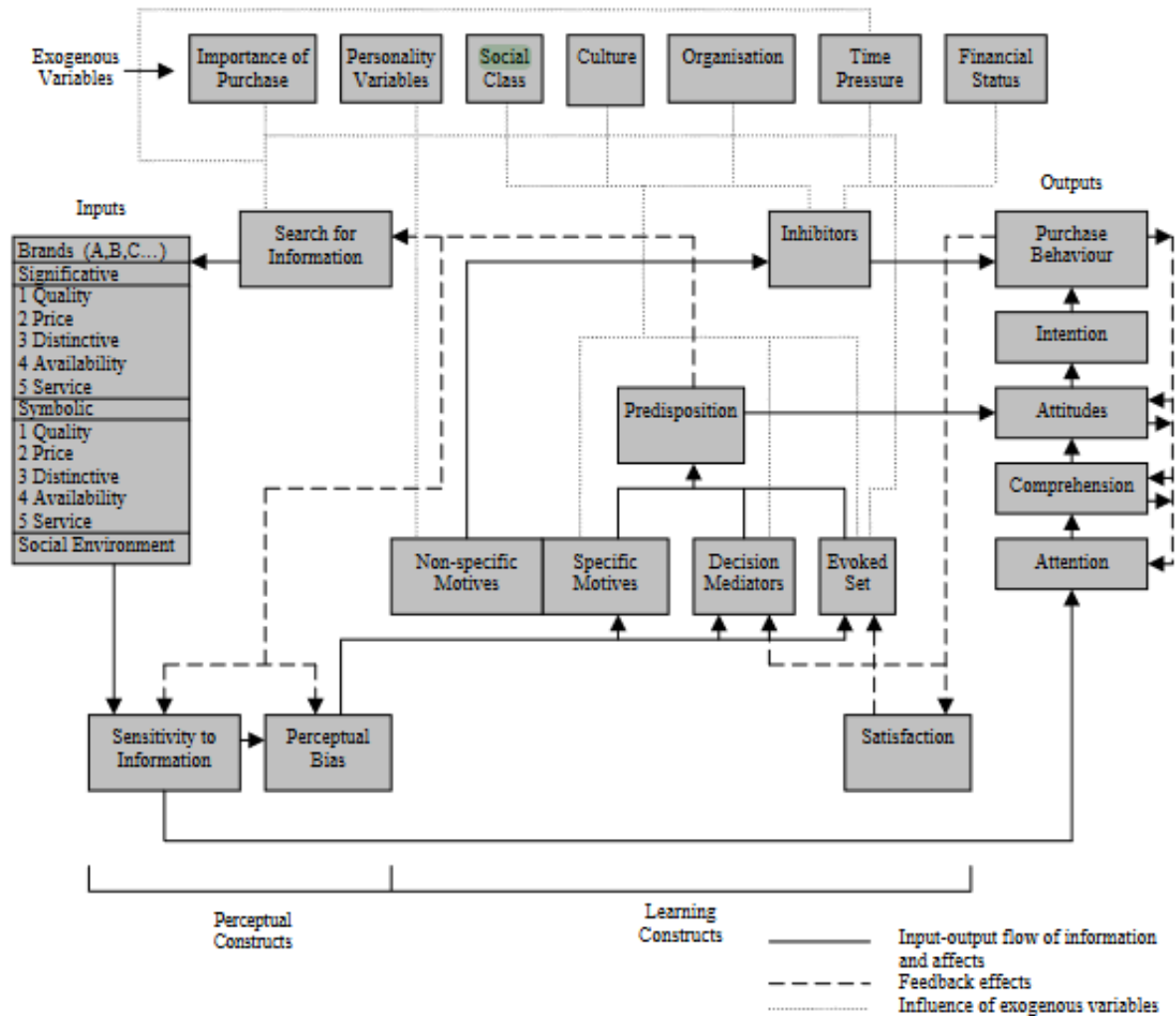
TABLE1: List Of Dataset Features and their Description

S/No	Feature Label	Feature Description	S/No	Feature Label	Feature Description
0	CUSTOMER_NO	Customer Purch. S/No	25	HOMEACC	Home Furniture
1	PURCHASE	Purchase (y/n)	26	LAMPS	Lamps Purch.
2	AMOUNT	Dollars Spent	27	LINENS	Linens Purch.
3	INCOME	Yearly Income	28	BLANKETS	Blankets Purch.
4	HOMEVAL	Home Value	29	TOWELS	Towels Purch.
5	FREQUENT	Order Frequency	30	OUTDOOR	Outdoor Prod.
6	RECENCY	Recency in Purch.	31	COATS	Coats Purch.
7	MARITAL	Married (y/n)	32	WCOAT	Ladies Coats
8	NTITLE	Name Prefix	33	WAPPAR	Ladies Apparel
9	AGE	Age	34	HHAPPAR	His/Her Apparel
10	TELIND	Telemarket Ind.	35	JEWELRY	Jewellery Purch.
11	APRTMNT	Rents Apartment	36	CUSTDATE	Date 1st Order
12	MOBILE	Occupied <1 yr	37	TMKTORD	Telemarket Ord.
13	DOMESTIC	Domestic Prod.	38	ACCTNUM	Account Number
14	APPAREL	Apparel Purch.	39	STATECOD	State Code
15	LEISURE	Leisure Prod.	40	RACE	Race
16	KITCHEN	Kitchen Prod.	41	HEAT	Heating Type
17	LUXURY	Luxury Items	42	NUMCARS	Number of Cars
18	PROMO7	Promo: 1-7 mon.	43	NUMKIDS	Number of Kids
19	PROMO13	Promo: 8-13 mon	44	TRAVTIME	Travel Time
20	COUNTY	County Code	45	EDLEVEL	Education Level
21	RETURN	Total Returns	46	JOB	Job Category
22	MENSWARE	Mens Apparel	47	VALRATIO	\$ Value per Mailing
23	FLATWARE	Flatware Purch.	48	DINING	Total Dining (kitch+dish+flat)
24	DISHES	Dishes Purch.	49	SEX	Sex

Note: Feature with S/No highlighted were considered in the analyses

APPENDIX B

Figure 1.4: The Theory of Buyer Behavior



Source: (Howard AND Sheth 1969, Loudon AND Della Bitta 1993)